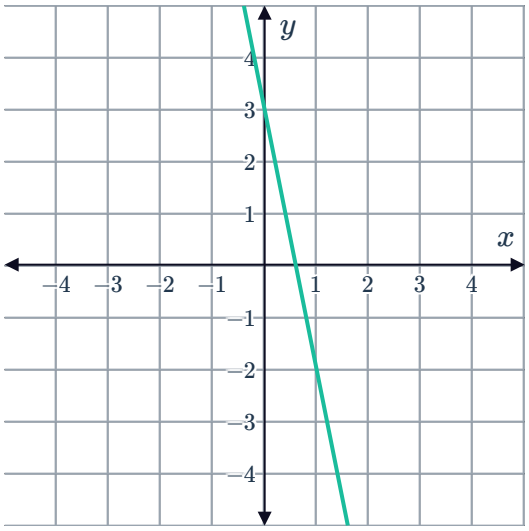
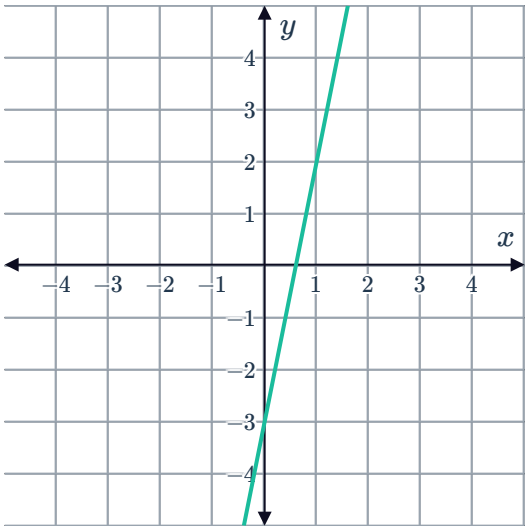
Characteristics from a graph

1 State whether each of the following graphs is a linear function decreasing from left to right.

a



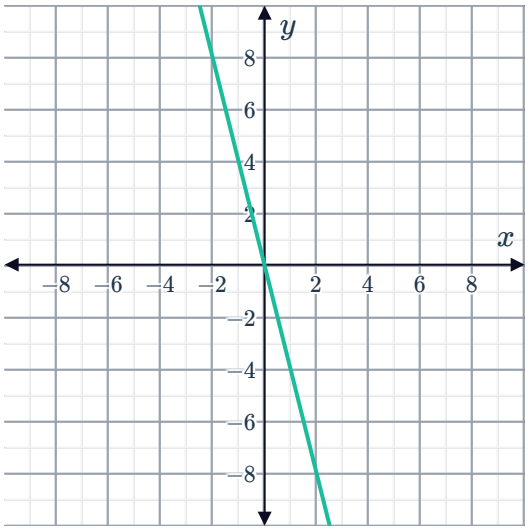
b



2 Consider the graph of $y = -4x$. Describe the end behavior of the y-values:

a As x becomes larger in the positive direction (i.e. x approaches ∞).

b As x becomes larger in the negative direction (i.e. x approaches $-\infty$).



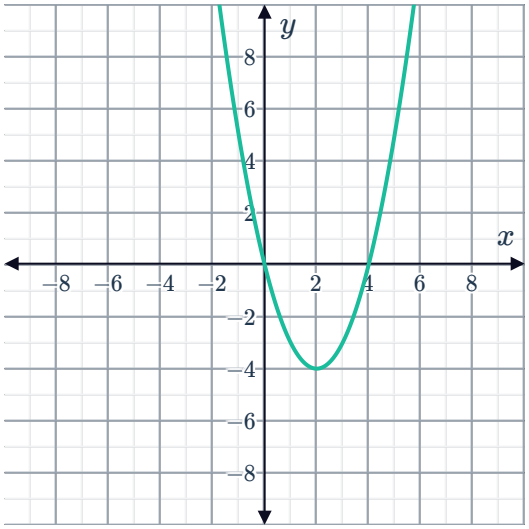
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3 Consider the following graphs.

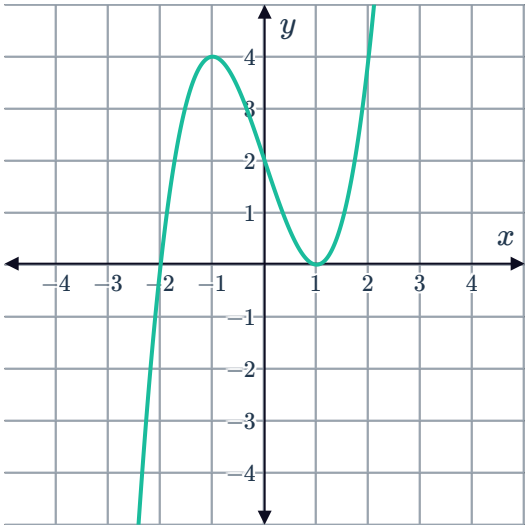
i State whether the function has an absolute maximum or minimum.

ii State whether the function has local maximums or minimums.

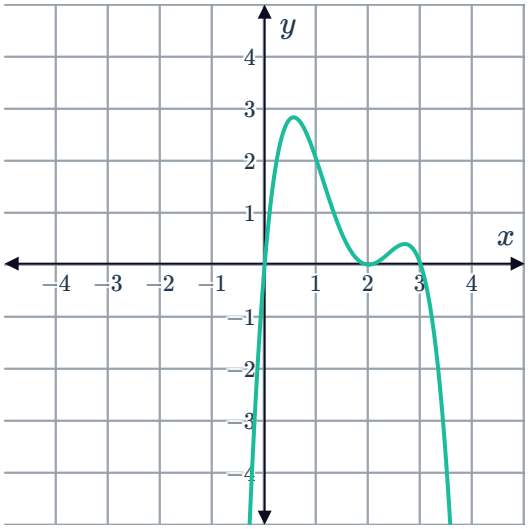
a



b



c



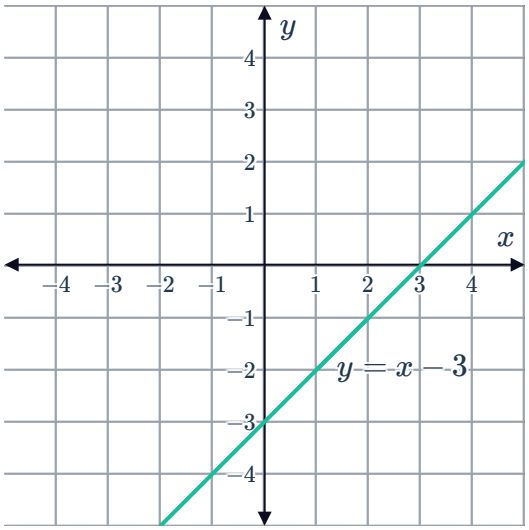
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4 Consider the following graphs:

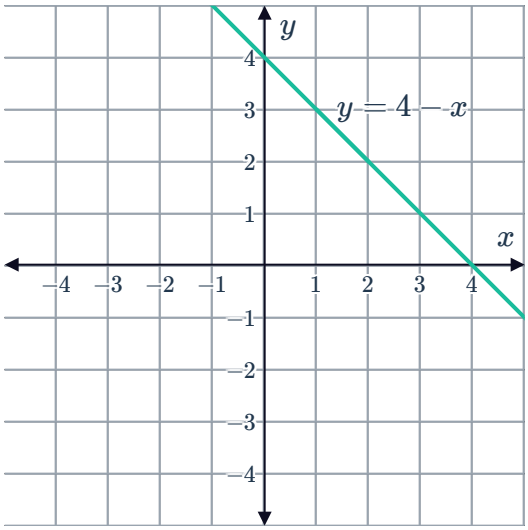
i State the coordinates of the x-intercept.

ii State the coordinates of the y-intercept.

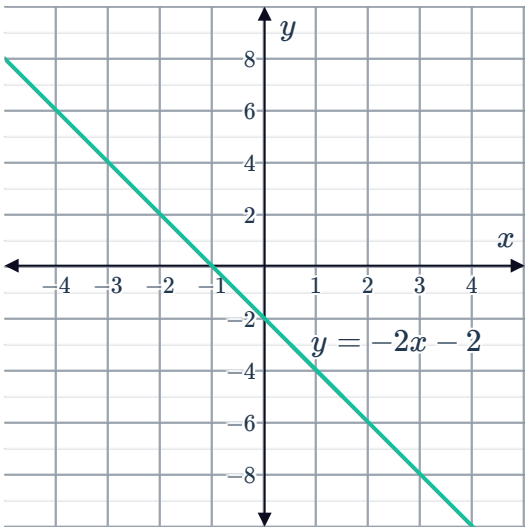
a



b



c



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5 Consider the functions shown on the following graphs:

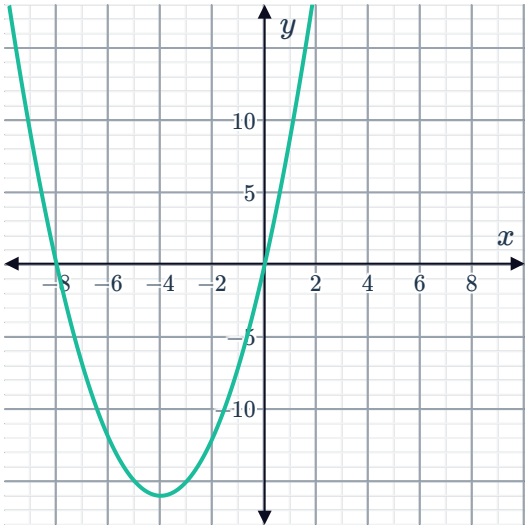
i State the coordinates of the x-intercepts.

ii State the coordinates of the y-intercept.

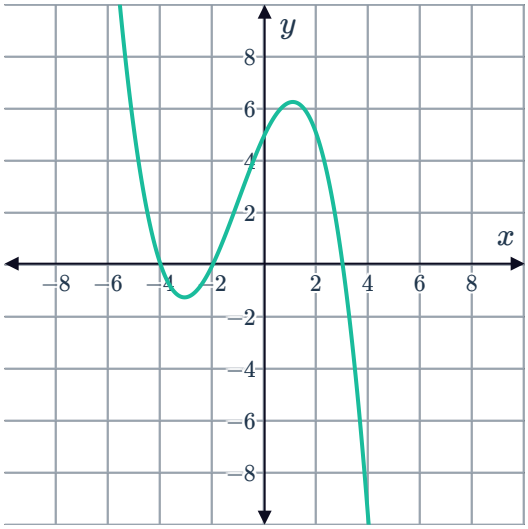
iii As $x \to \infty$, describe the end behavior of the corresponding y-values.

iv As $x \to -\infty$, describe the end behavior of the corresponding y-values.

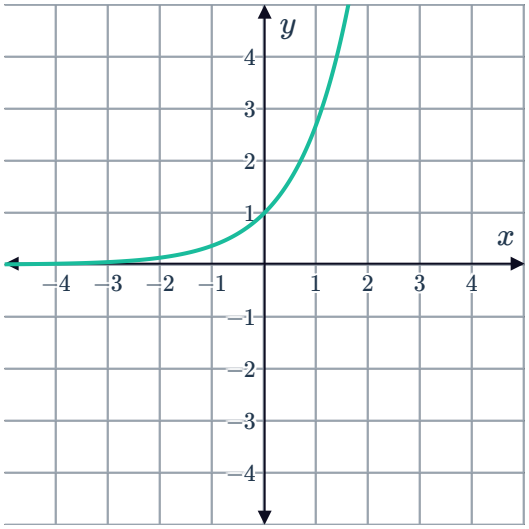
a



b

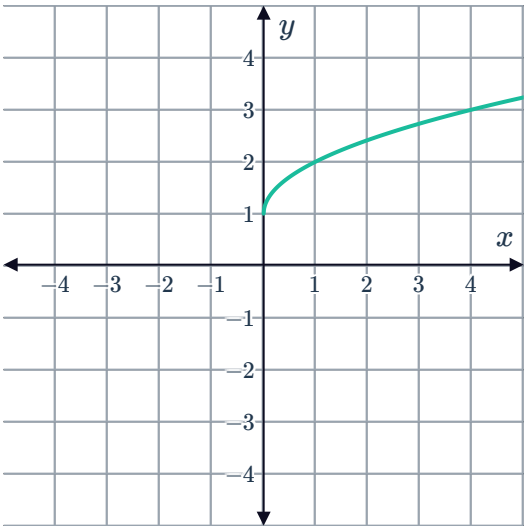


c



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6 Consider the graph below.

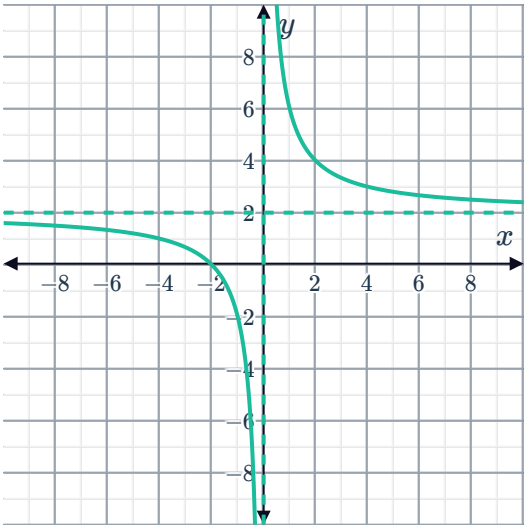


a State the coordinates of the x-intercept.

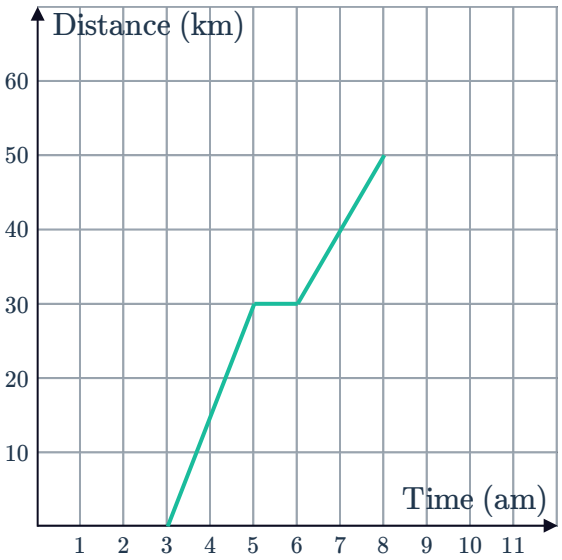
b State the coordinates of the y-intercept.

c As x becomes larger in the positive direction (ie. x approaches ∞), describe the end behavior of the y-values.

7 State the coordinates of the x-intercept of the following graph:



8 The graph shown describes Neil's distance from home throughout his journey.



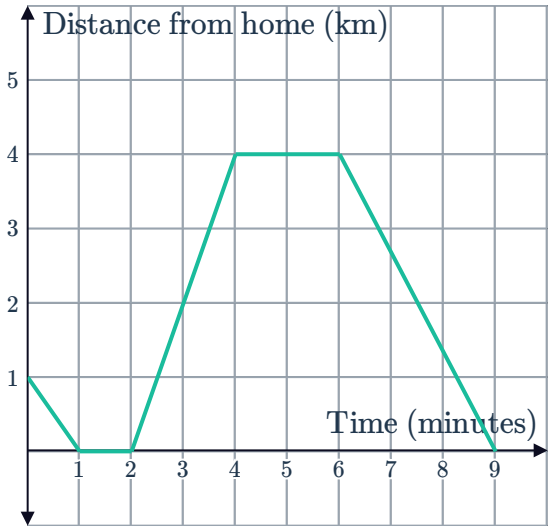
a How far was Neil from home when he started driving at 3 am?

b How far was he from home by 5 am?

c How long did he take to rest?

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9 The graph represents Judy's distance from home over a 9 minute interval.
Determine Judy's initial distance from home.

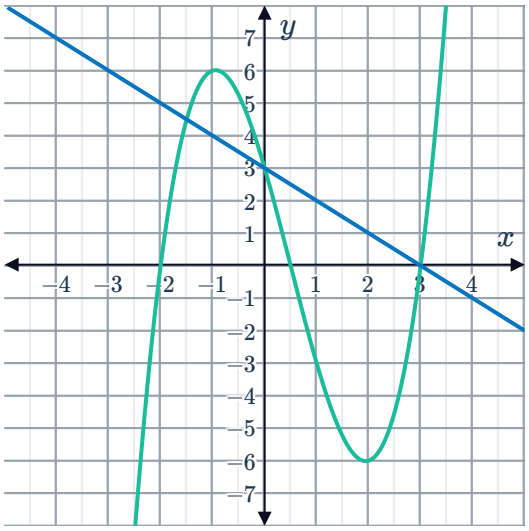


10 Consider the function $f(x) = -x^2 + 3x - 2$.
Determine the interval for which interval the function is both decreasing and positive.

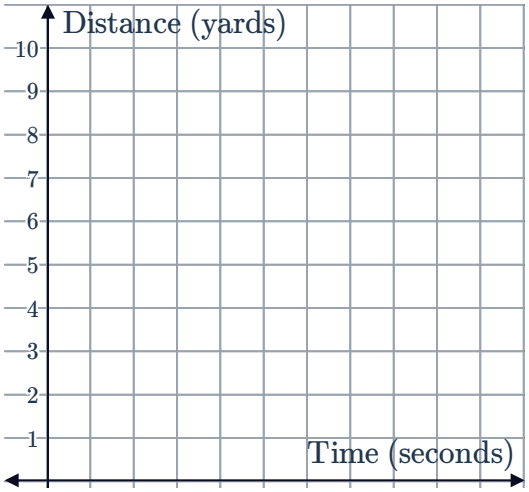
11 Consider the functions $f(x)$ and $g(x)$ shown below.
Determine whether $f(x) \neq g(x)$ for each of the following values of x.

a 0 **b** 3

c -3 **d** $-\frac{3}{2}$



12 A puppy is playing outside. Observe the following graph which shows the puppies distance $d(t)$, in yards from home over a period of time, t, in seconds.
Determine the interval representing the puppy constantly moving further from home.



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13 The amount of water (million gallons per hour) flowing into a dam is recorded. For the first 4 hours, it flowed at an average rate of 1 million gallons per hour. The water then started to flow at an average rate of 0.5 million gallons per hour for the next 3 hours. Then it stopped flowing for 2 hours. Then it started flowing again at an average rate of 1.5 million gallons per hour for the next 3 hours.

a Graph the function showing the accumulated water flow over the recorded period.

b Determine the average rate of flow over the length of the recorded period.

Additional questions

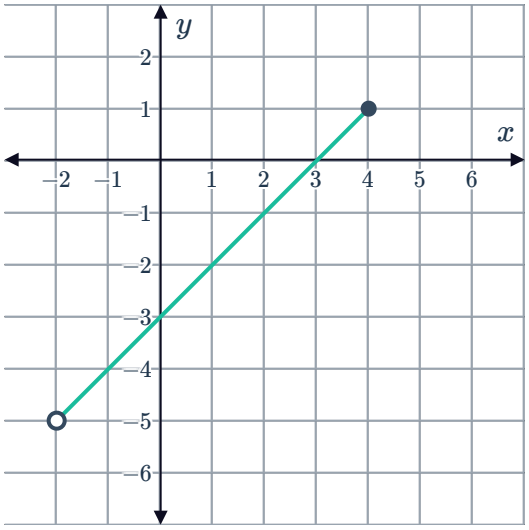
14 For each of the following graphs, state the:

i Domain

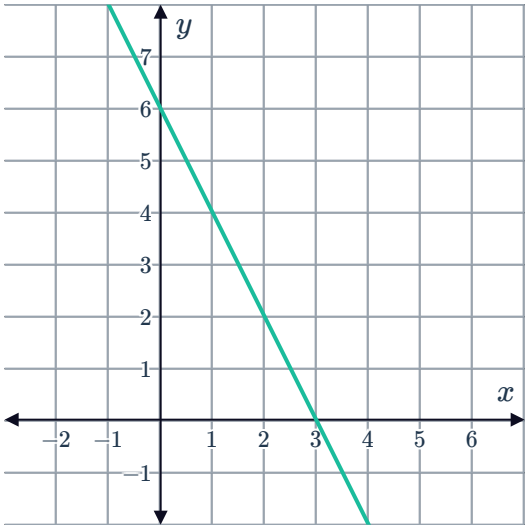
ii Range

iii x- and y-intercepts

a

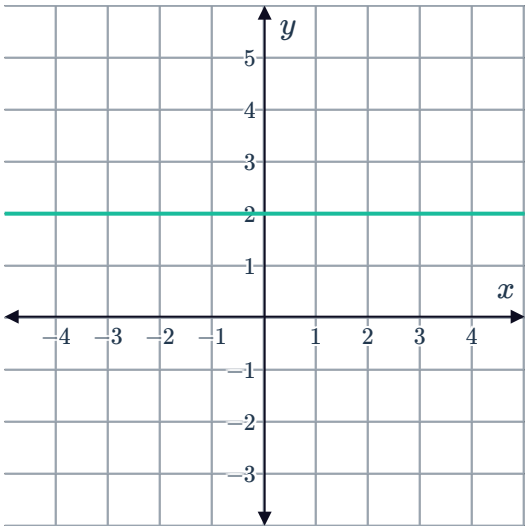


b

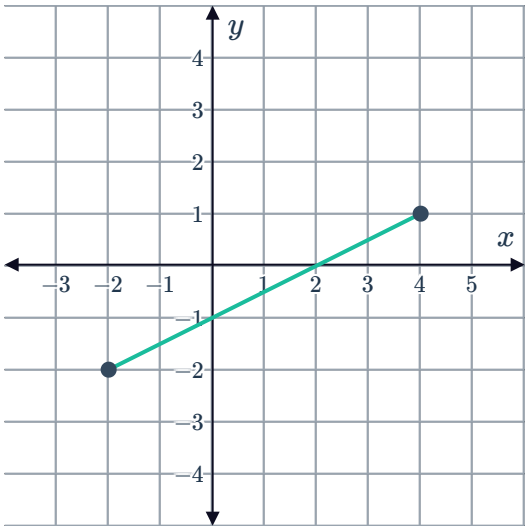


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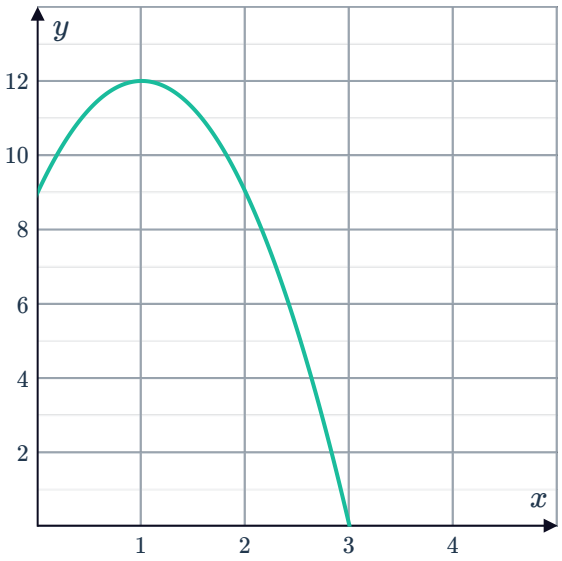
c



d



15 The graph shows the height of a soccer ball against the time that has passed after it has been kicked.



a Find the coordinates of the y-intercept.

b Interpret the y-value of the y-intercept in this context.

c Find the coordinates of the x-intercept.

d Interpret the x-value of x-intercept in this context.

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